Chest X-Ray Generation with a VAE

ECE 285: Deep Generative Models — Homework 1

Abstract

A convolutional VAE trained on 5,216 chest X-rays generates blurred but recognizable thoracic structure. Validation selects epoch 45 from 50 epochs. Comparing 624 prior samples with 624 held-out images gives FID 283.72 and Inception Score 2.095 using ImageNet Inception-v3. These scores describe appearance rather than clinical fidelity.

1 Data and model

The Kaggle Chest X-Ray Images (Pneumonia) dataset [2] provides 5,216 training, 16 validation, and 624 test images. We preserve these splits, ignore labels, and convert images to 128×128 grayscale in $[0, 1]$ without augmentation. Test images are reserved for final evaluation.

Four 4×4 stride-2 convolutions (32/64/128/256 channels, ReLU) encode each image. Two linear heads define a 128-D diagonal Gaussian; a linear projection and four transposed convolutions decode it. The model has 7,686,081 parameters. VAE reparameterization [1] uses $z = \mu + \exp(\frac{1}{2} \log \sigma^2) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. Training minimizes

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\mathrm{BCE}(x, \sigma(f_\theta(z)))] + \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \| \mathcal{N}(0, I)). \tag{1}$$

BCE is summed over pixels and KL over latent dimensions, then averaged over images. BCE on continuous intensities is a reconstruction surrogate.

Optimizer	Adam	Learning rate	10^{-3}
Batch size	64	Epochs	50
Latent size	128	KL weight β	1
Training seed	42	Adam (β_1, β_2)	$(0.9, 0.999)$

2 Training behavior

Training loss falls from 10,411.04 to 9,431.28 nats/image. Validation selects epoch 45 (loss 9,059.32) for generation. KL remains nonzero.

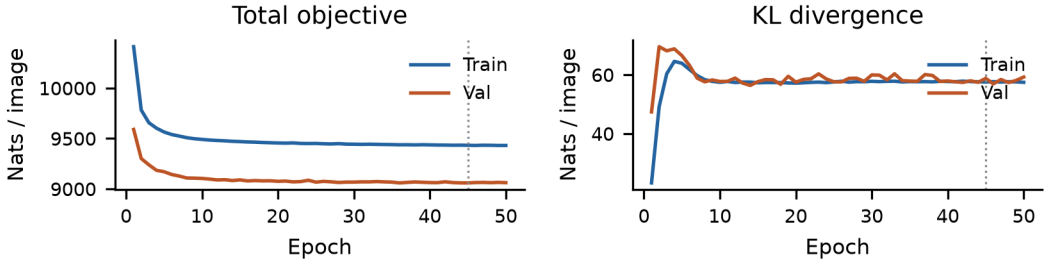


Figure 1: Training and validation losses. Dotted line: selected epoch 45.

3 Generation and evaluation

For the selected checkpoint, we draw 624 samples from $z \sim \mathcal{N}(0, I)$ with CPU seed 2026 and batch size 32. These are prior samples, not reconstructions. Figure 2 shows the first 16 draws without visual selection.

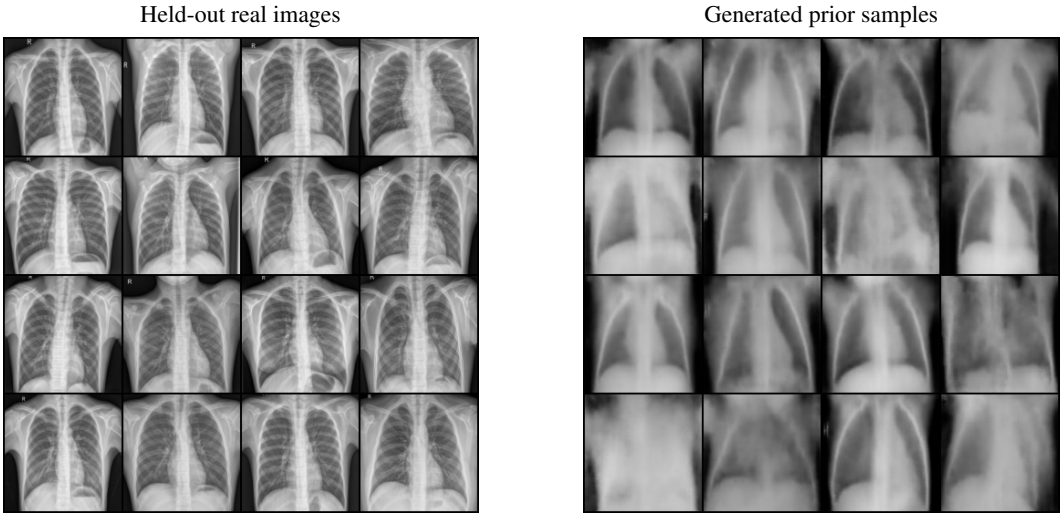


Figure 2: Left: first 16 sorted test images (all NORMAL). Right: first 16 prior samples. Metrics use all 624 test images, including both classes.

Following the assigned Ignite tutorial [3], both sets use 8-bit PNGs, three identical channels, and PIL bilinear resizing to 299×299. We apply ImageNet mean/std normalization. Torchvision's ImageNet Inception-v3 supplies 2,048-D average-pool features to Ignite FID and 1,000-class softmax probabilities to Ignite IS. Pooled features replace Ignite's default classifier outputs; this differs from the TensorFlow-compatible `pytorch-fid` protocol.

Metric (624 real / 624 generated images)	Measured value
FID, pooled 2,048-D features ↓	283.72
Inception Score, generated, full set ↑	2.095
Inception Score, generated, 10 splits (mean ± SD)	2.045 ± 0.188
Inception Score, real test, full set (context only)	2.284

4 Discussion and limitations

Samples show lung outlines, a bright mediastinal region, and varying chest shape, but blurred ribs and boundaries. Reduced loss supports learning coarse appearance, not diagnostic fidelity. The small, balanced validation split differs from the training set (1,341 NORMAL / 3,875 PNEUMONIA), possibly contributing to its lower loss. FID [4] has singular 2,048-D sample covariances at $n = 624$. IS uses ImageNet classes, not medical labels, and has known limitations [5]. Neither establishes clinical correctness. Split SD is not a confidence interval; one training run cannot establish robustness. Sharper decoders and domain-specific evaluation are useful next steps.

References

[1] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.
[2] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle. Dataset page.
[3] PyTorch-Ignite. GAN evaluation using FID and IS.
[4] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.
[5] S. Barratt and R. Sharma. A Note on the Inception Score. arXiv:1801.01973, 2018.